MOSAIC: Matched-Observation Study of Architecture-Informed Control for Hybrid QKD Routing

Author Names Withheld for Draft

***Abstract*—We study simulator-based next-hop routing in a modeled trusted-node quantum key distribution network with fiber and free-space optical links. We compare graph neural network proximal policy optimization (GNN-PPO) with a matched long short-term memory policy (LSTM-PPO). Both share the actor–critic head, candidate-edge features, training budget, and physical-feasibility mask; five independently trained checkpoints per policy are tested on identical held-out scenarios. Route completion was 46.1% for GNN-PPO and 74.6% for LSTM-PPO, a paired difference of 28.5 percentage points (nested 95% interval: 20.6–37.2). The LSTM implementation encodes node-feature rows in a fixed node order and resets its hidden state on each decision, so it is not a persistent temporal-memory baseline. These simulator results use proxy topology and link quantities, not verified infrastructure, device measurements, or delivered secret-key throughput.**

***Index Terms*—Quantum key distribution, trusted-node networks, routing, graph neural networks, attention, proximal policy optimization, free-space optical communication.**

1. INTRODUCTION

Trusted-node quantum key distribution (QKD) networks extend point-to-point key generation by relaying keys through intermediate nodes. A route determines which links must be available and which stored key resources are consumed, so network-level delivery depends on channel state and resource availability as well as geometric distance [1]–[4]. Routing therefore links physical conditions to end-to-end service feasibility.

This decision is especially state-dependent in hybrid fiber and free-space optical (FSO) networks. Fiber attenuation varies with span length and device parameters, while FSO links also depend on atmospheric attenuation, turbulence, pointing, and availability. Link error, usable key-rate state, stored key pools, and time-varying availability can change which next hop is feasible and desirable [5], [6]. We ask: *Does graph message passing improve constrained next-hop routing over a matched LSTM-PPO node-sequence encoder in a hybrid fiber/FSO trusted-node QKD network?* We study one route per episode, rather than traffic-level routing, wavelength/time-slot assignment, or demand blocking.

Prior QKD-routing studies address related but distinct decisions. Sharma *et al.* jointly optimize routing and optical-resource assignment for QKD lightpath requests under traffic load [7]; Johann *et al.* study route selection in meshed trusted-node networks with key-store state and network-level requests [6]. Learning-based routing, including PPO-based entanglement routing, is also established, so reinforcement learning itself is not the contribution claimed here [8], [9]. The present study instead tests graph-structured sequential next-hop decisions with fiber/FSO channel and normalized resource state in the observation, against a matched ordered-node LSTM policy.

We contribute (1) a hybrid fiber/FSO trusted-node simulator coupling scenario-dependent channel and normalized key-pool state to routing observations; (2) a matched GNN-PPO/LSTM-PPO comparison using five independent training seeds and paired held-out scenarios; and (3) route-completion, route-length, condition, seed-variability, inference-cost, and reward-component analyses. Across the frozen 540-scenario held-out matrix, LSTM-PPO completed 74.6% of routes versus 46.1% for GNN-PPO, a paired difference of 28.5 percentage points (nested 95% interval: 20.6–37.2). This answers the stated question in favor of the tested LSTM implementation under this simulator; its fixed node order and per-decision state reset limit that interpretation. The result does not establish field performance or secret-key service capacity. Section II reviews related work; Section III describes the simulator and protocol; Section IV formulates the routing policy and comparator; Section V reports results; Section VI discusses interpretation and limits; and Section VII concludes.

2. BACKGROUND AND RELATED WORK

A. QKD and Trusted-Node Networking

QKD protocols combine a quantum channel with authenticated classical communication. BB84 uses nonorthogonal signal states to allow the communicating parties to estimate disturbance and distill key material [10]. Practical transmitters commonly use weak coherent pulses rather than ideal single-photon sources. Decoy-state methods use pulses with different intensities to estimate single-photon contributions and counter photon-number-splitting attacks [11], [12]. Finite-key operation must include statistical fluctuations, error-correction leakage, and security parameters. Asymptotic rate formulas are useful for link-level screening, but they do not by themselves certify a finite block. Entropic uncertainty and composable security analyses provide established foundations for finite-size bounds [13]–[15]. For STN chains, the parties and parity announcements couple information across links. A finite-key bound for such a chain is therefore protocol-specific and can-

This manuscript is a working draft based on the QKD routing simulator maintained by the authors. Author affiliations, funding, and correspondence details are to be completed.

not be replaced by summing independent per-link asymptotic rates [16].

In a conventional trusted-node chain, neighboring QKD modules establish link keys. The network forwards end-to-end key material through trusted relays, commonly using one-time-pad operations. For a path with link key rates R_i, the instantaneous end-to-end service rate is limited by the slowest required resource, and practical delivery also depends on stored key inventory and demand. Key pools therefore couple routing decisions over time [4], [6].

B. Fiber and FSO QKD Links

Long fiber spans motivate intermediate trusted nodes. FSO links can provide candidate bypasses where a direct fiber span is unavailable or degraded. Their channel quality depends on atmospheric attenuation, turbulence, pointing, and availability. Terrestrial FSO design methods identify these as separate budget terms and require local environmental information for credible planning [17], [18]. The simulator models time-of-day and seasonal scenario variables, but its current FSO distributions are assumptions rather than fits to site-specific Indian weather traces.

The routing quantity used in this work is a normalized simulator rate proxy, not an experimentally validated secret-key throughput. The physical model maps attenuation, detector efficiency, and dark-count settings to detection and QBER estimates; its optional decoy-state finite-key engineering mode is not a composable security proof. FSO weather and turbulence distributions are synthetic scenarios rather than site-calibrated meteorological traces.

C. QKD Routing and Resource Management

Classical shortest-path methods provide reproducible baselines but optimize a fixed edge cost or hop count [19]. QKD network research has therefore studied key-store use, request blocking, and route/resource assignment. Sharma *et al.* use deep reinforcement learning to jointly solve routing and resource assignment for QKD lightpath requests in optical networks; their action includes optical resource allocation and their metrics include blocking and resource utilization under traffic [7]. Johann *et al.* formulate meshed trusted-node route selection around key-store state and network-level requests [6]. These formulations address multi-request service and resource assignment, while the present simulator isolates constrained next-hop selection for a single route and does not model wavelength/time-slot assignment or demand blocking. Other learning-based work studies entanglement routing or mixed trusted-node/repeater settings [8], [9]; these differ in network resources and route semantics.

D. Graph Reinforcement Learning

Sequential decision methods can represent state-dependent link availability; dynamic programming provides a classical foundation for sequential optimization [20]. PPO is a clipped policy-gradient method, and generalized advantage estimation reduces variance in its return estimates [21], [22]. The partially observed formulation follows standard POMDP and reinforcement-learning foundations [23], [24]. Graph neural networks encode neighborhoods by message passing, for example

$$h_v^{(\ell+1)} = \phi\Big(h_v^{(\ell)}, \mathrm{AGG}_{u\in\mathcal{N}(v)}\, \psi(h_v^{(\ell)}, h_u^{(\ell)}, e_{uv})\Big). \quad (1)$$

Here, $h_v^{(\ell)}$ is the embedding of node v at layer ℓ, e_{uv} is the edge feature, ψ is a learned message map, AGG combines incoming messages, and ϕ updates the node embedding. Graph convolution, neighborhood aggregation, and attention are established choices for these operators [25]–[27]. This representation is relevant when candidate actions are neighboring nodes and link state must be combined with route context. Whether it improves decisions over a matched non-graph policy remains an empirical question.

3. SIMULATOR ARCHITECTURE AND EXPERIMENTAL SETUP

A. Architecture

The simulator couples topology and link profiles to a time-stepped environment, routing policy, and evaluation layer. At each step, scenario variables update fiber/FSO link states; the environment forms graph and candidate-edge observations, applies a shared feasibility mask, and updates route and normalized key-pool state after the selected hop. Fig. 1 summarizes this physical-to-decision coupling; the nine candidate-edge features are defined in Table II. The controller is a centralized simulator abstraction, not a deployed QKD management interface.

B. Simulation Setup

1. Topology and Corridor Data

The topology contains seven city hubs (Delhi, Jaipur, Mumbai, Bangalore, Hyderabad, Kolkata, and Chennai), 2,370 modeled nodes, and 2,653 links: 273 fiber links and 2,380 candidate FSO links. Fig. 2 distinguishes the modeled link classes and marks the example request endpoints. OpenStreetMap driving routes provide corridor geometry for the modeled fiber paths; this is a planning proxy, not verified deployed fiber infrastructure. FSO line of sight and terrain clearance are not surveyed. The topology is archived with the research data and uses OpenStreetMap data under its applicable attribution and database terms [28].

2. Link and Noise Models

Fiber loss is represented using wavelength-dependent attenuation and excess loss. Detector efficiency, dark-count bounds, and a visibility assumption contribute to detection and QBER proxies. The selected fiber device profile is based on published/vendor-reported ID Quantique ID281 values at 1550 nm. The FSO channel is a separate assumed 785-nm scenario with its own efficiency and dark-count settings. These are not a single matched commercial optical terminal profile.

FSO turbulence is represented with a correlated latent log-C_n^2 process. The environment samples availability and channel conditions by time step; FSO corridor outages are held fixed for the episode to avoid a bypass disappearing between its relay segments. These choices create repeatable stress scenarios. A detailed parameter ledger and provenance categories are supplied in `PHYSICS_EVIDENCE.md`.

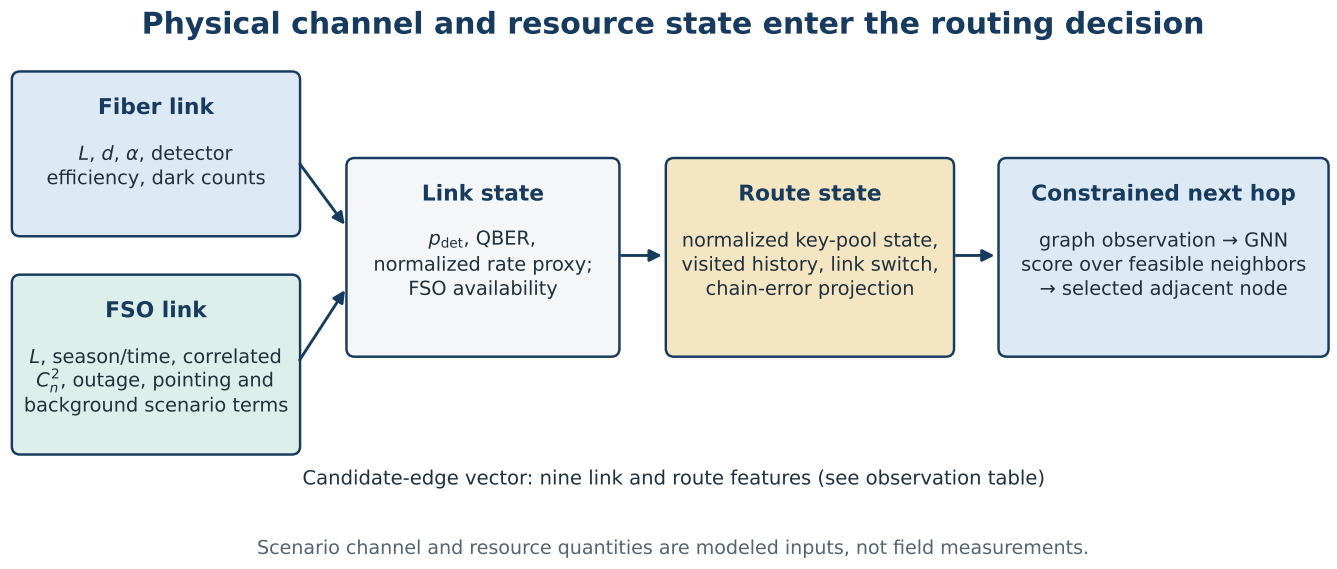


Fig. 1. Physical channel and normalized resource state flow into constrained next-hop selection. Fiber and FSO inputs determine link state; route state and nine candidate features inform the feasibility-masked policy. Channel and resource quantities are simulator inputs.

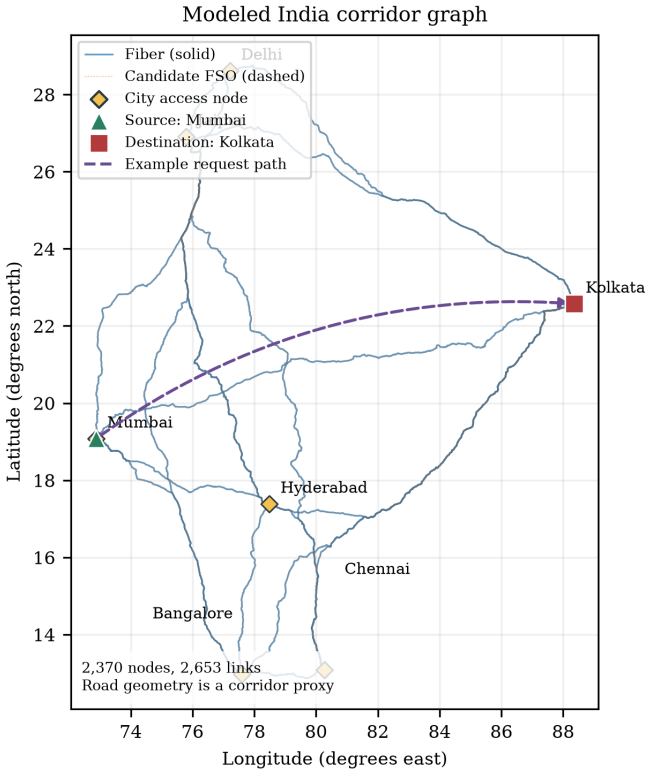


Fig. 2. Modeled corridor graph: solid blue lines denote fiber, dashed orange lines denote candidate FSO links, diamonds mark city access nodes, and distinct symbols mark the Mumbai source and Kolkata destination. The purple dashed route is an example request, not a physical link. Road geometry is a corridor proxy; FSO line of sight is assumed.

TABLE I
REPRESENTATIVE LINK-MODEL PARAMETERS AND EVIDENCE CLASS

Quantity	Configured value	Evidence class
Fiber wavelength	1550 nm	device profile
Fiber attenuation	0.20 dB/km	inherited target
Fiber detector efficiency	0.93	measured device value
Fiber dark-count bound	$< 70\ \mathrm{s}^{-1}$	measured upper bound
FSO wavelength	785 nm	scenario assumption
FSO detector efficiency	0.60	scenario assumption
FSO dark counts	$100\ \mathrm{s}^{-1}$	scenario assumption
FSO temporal correlation	0.85	uncalibrated assumption

The implemented FSO model uses a correlated log-normal turbulence process with atmospheric, diffraction, pointing, and background terms. Detector and channel noise enter through the QBER and normalized rate calculations. The optional finite-key estimator is an engineering model, not an independently verified composable proof.

The default rate mode is an asymptotic, normalized proxy of the form

$$r_{\mathrm{proxy}} = p_{\mathrm{det}}\left[1 - 2h_2(Q)\right], \tag{2}$$

where p_{det} is modeled detection probability, Q is modeled QBER, and $h_2(Q) = -Q\log_2 Q - (1-Q)\log_2(1-Q)$ is binary entropy, with endpoint values defined by continuity. This asymptotic expression is used as a normalized simulator proxy. An opt-in decoy-state mode estimates finite-key contributions and uncertainty; that engineering estimate is not a composable security proof.

3. *Environment and Protected Conditions*

The observation uses relative node features and candidate-edge features; absolute coordinates and node identifiers are excluded in the evaluated configuration. Candidate actions are

TABLE II
OBSERVATION FEATURES FOR EACH CURRENT NODE AND CANDIDATE EDGE

Feature group	Components
Node (5)	City indicator; full-TN indicator; STN indicator; destination-relative distance; normalized degree.
Candidate edge (9)	QBER; normalized rate proxy; FSO indicator; normalized distance; projected chain error; normalized key-pool state; link-type switch; visited-node flag; destination-distance progress.
Message-passing edge (4)	QBER; normalized rate proxy; FSO indicator; normalized distance.
Feasibility	Adjacent node; QBER below threshold; positive rate proxy; projected chain error below threshold.

adjacent nodes only. Within each untrusted segment, the model assumes independent hop errors Q_i. Their cumulative parity-error probability after k hops is

$$Q_{\mathrm{chain},k} = \frac{1 - \prod_{i=1}^{k}(1 - 2Q_i)}{2}, \qquad Q_{\mathrm{chain},0} = 0. \tag{3}$$

The accumulator resets to zero on arrival at a full trusted node. Let $\mathcal{N}(v_t)$ be the neighbors of the current node and define $m_{t,u} = 1$ only if edge (v_t, u) has QBER below $Q_{\max}$, positive normalized rate proxy, and projected segment error from (3) below $Q_{\max}$. The feasible action set is

$$\mathcal{A}_{\mathrm{feasible}}(s_t) = \{u \in \mathcal{N}(v_t) : m_{t,u} = 1\}, \tag{4}$$

with $Q_{\max} = 0.11$. This mask enforces feasibility; the policy's utility is measured by its selection among the remaining feasible neighbors.

4. Reward Definition

Feasibility and utility are distinct: the mask in (4) removes invalid actions, while the shaped reward ranks feasible moves. For the defence profile used by the archived experiments, the per-step return is the sum of link, route, and terminal terms:

$$\begin{aligned} r_t = {} & w_p R_{p,t} + w_q R_{q,t} + w_r R_{r,t} + w_k (R^{\mathrm{level}}_{k,t} + R^{\mathrm{delta}}_{k,t}) \\ & - C_{\ell,t} - C_{e,t} - C_{c,t} + C_{\mathrm{step},t} + C_{\mathrm{revisit},t} + C_{\mathrm{switch},t} \\ & + C_{\mathrm{backtrack},t} + C_{\mathrm{arrival},t} + C_{\mathrm{security},t} + C_{\mathrm{depletion},t}. \end{aligned} \tag{5}$$

The progress, QBER-margin, rate, and pool terms are signed by destination-distance progress; link costs and terminal penalties are applied as implemented. Table III gives the defence-profile weights and fixed protection values used in the current reward audit.

5. Training and Evaluation Setup

The comparison uses five independently trained GNN-PPO and five LSTM-PPO policies, matched behavior-cloning and PPO budgets, endpoint exclusion from both training stages, a monsoon-only validation suite, and identical held-out endpoint/season/time scenarios. The protocol and resolved per-seed configurations are supplied in the supplementary package. Checkpoint selection uses validation scenarios only; held-out test scenarios are reserved for final reporting.

The LSTM-PPO comparator shares the GNN policy's actor-critic head, candidate-edge features, PPO objective, and physical-feasibility mask. It replaces message passing with a single-layer LSTM over node-feature rows in the topology's stable node order and receives no graph edges. The LSTM state is initialized afresh for each policy call, so this is a node-sequence encoder rather than persistent memory across route decisions. Matching the remaining components and training budgets makes encoder type the primary architectural difference under test.

TABLE III
REWARD TERMS FOR THE DEFENCE PROFILE

Term	Weight/value	Per-step contribution
Progress	0.2	Destination-distance reduction, normalized by max edge distance.
QBER margin	0.8	Feasible projected-chain margin, signed by progress.
Rate proxy	1.2	Distance-prorated normalized link-rate utility, signed by progress.
Pool level / change	0.4 each	Current normalized pool and its change, signed by progress.
Latency / energy	0.4 each	Distance-based costs with configured normalization constants.
Congestion	0.5	Degree cost above two incident choices at the departure node.
Step / revisit	-0.05 / -0.6	Fixed step and revisit costs.
Switch / backtrack	-0.2 / -0.5	Link-type switch and non-progress cost.
Arrival	$+4.0$	Destination terminal bonus.
Security / depletion	-30 / -60	Terminal penalties for chain-error breach and pool depletion.

TABLE IV
FROZEN PROTOCOL FOR THE FINAL MATCHED ARCHITECTURE AND HOLDOUT STUDY

Setting	Value
PPO endpoint and training season	Delhi–Chennai; monsoon, 22:00
BC demonstrations	Random pairs excluding test pairs and reversals
Independent training seeds	20261002–20261006 per architecture
Matched architectures	GNN-PPO and LSTM-PPO
Training budget	15 BC epochs; 50 PPO epochs; 144 steps/epoch
Validation	Delhi–Chennai; monsoon; seeds 11001–11004
Held-out endpoints	Six ordered pairs; reversals excluded from BC
Held-out environment	Normal, summer, winter; 02:00 and 22:00
Paired test suite	108 shared scenarios per checkpoint
QBER / outage / episode limit	0.11 / 0.01 / 144 steps

4. ROUTING POLICY AND LEARNING OBJECTIVE

A. Graph State and Constrained Routing

At time t, let $G_t = (V, E, X_t)$ denote the modeled graph and its scenario-dependent link state, where X_t contains channel and normalized pool values. The environment state s_t additionally contains current and destination nodes, route history, chain-error state, and time. The policy receives observation o_t and selects an adjacent node $a_t \in \mathcal{A}_{\mathrm{feasible}}(s_t)$ as defined in (4). Transitions update the route, key-pool level, and channel scenario; future turbulence and outage draws are not exposed as a forecast to the policy.

An episode succeeds upon reaching the destination. It terminates as a failure after a projected security-bound breach or modeled pool depletion, and is truncated at the configured step limit if neither terminal condition has occurred. If no

feasible neighbor remains, the policy has no selectable action and the episode is recorded as a no-valid-action failure by the evaluator. These conditions define route completion within this simulator and are not a cryptographic security certification.

The objective is to find a policy $\pi_\theta(a_t \mid o_t)$ that maximizes expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \gamma^t r_t\right], \tag{6}$$

subject to the action constraint in (4). The reward in (5) is a training signal; evaluation reports route success, successful-route hop count, invalid-action/fallback counts, and condition and seed variation.

1. Graph Policy and Action Mask

For each node v, the message-passing stack computes an embedding h_v from node features and graph connectivity. The encoder uses QBER and positive rate proxy to filter invalid full-graph edges; the message function itself aggregates learned transformations of endpoint embeddings. The actor's attention head then combines current, destination, and candidate-neighbor embeddings with the candidate's nine route-context features. It produces logits $z_{v,u}$ for adjacent candidates. Invalid candidates are assigned negative infinity before softmax:

$$\pi_\theta(u \mid o_t) = \frac{\exp(z_{v_t,u})}{\sum_{w \in \mathcal{A}_{\text{feasible}}(s_t)} \exp(z_{v_t,w})}, \quad u \in \mathcal{A}_{\text{feasible}}(s_t). \tag{7}$$

The distribution is normalized over the feasible set defined in (4). Invalid links contribute no message in either direction, and the candidate mask additionally enforces projected chain error. If this set is empty, the evaluator records a no-valid-action failure rather than sampling from an empty distribution. This separates the environment's feasibility filter from the learned preference: the policy chooses among feasible alternatives.

In the optional attention variant, candidate embeddings are combined using learned compatibility scores, following neighborhood attention formulations [27]. Attention is treated as an architectural choice. Its value must be established by an ablation with the same training budget and seed distribution. The current results do not isolate a causal benefit from attention.

2. PPO Training

The policy is updated by maximizing the clipped surrogate objective

$$L^{\text{clip}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t(\theta)\hat{A}_t, \; \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\Big)\Big], \tag{8}$$

where $\rho_t(\theta) = \pi_\theta(a_t \mid o_t)/\pi_{\theta_{\text{old}}}(a_t \mid o_t)$ is the new-to-old action-probability ratio, $\hat{A}_t$ is the estimated advantage, and ϵ is the clipping threshold. PPO is combined with a value loss and entropy regularization. Hyperparameters are resolved from a versioned YAML file and included with the experiment artifact. The settings for the transfer-evaluation checkpoint and paired episode seeds are archived in the supplementary package.

B. Encoder-Level Comparison

Fig. 3 summarizes the shared and encoder-specific components.

Both encoders feed node representations to the same candidate scorer, critic, and masked action distribution. GNN-PPO aggregates information over the graph adjacency at each message-passing layer. LSTM-PPO instead processes the node-feature matrix as a sequence in the topology's stable node order, without edge-index input, using the standard recurrent LSTM encoder [29]. For sequence position i, its encoder computes

$$(h_i, c_i) = \text{LSTM}(x_i, h_{i-1}, c_{i-1}), \qquad h_0 = c_0 = 0, \tag{9}$$

and supplies h_i to the shared candidate scorer. The hidden and cell states are initialized afresh for each policy call. Thus, the LSTM can mix node features according to their sequence positions, but it has no persistent state across route decisions and no explicit graph adjacency. This comparison tests the implemented message-passing encoder against the implemented ordered-node sequence encoder under a shared decision head. Sensitivity to the arbitrary node order is therefore a limitation of the comparator and of the interpretation.

5. RESULTS

A. Matched GNN-PPO and LSTM-PPO Held-Out Comparison

The frozen protocol in `supplementary/final_gnn_lstm_protocol_20261007.md` evaluates five independently trained checkpoints per architecture on six ordered city pairs, six season/time conditions, and three shared environment replicates (108 matched scenarios per checkpoint; 540 episodes per architecture). Training seeds and environment seeds are paired across encoders. We summarize uncertainty by resampling training checkpoints and, within each selected checkpoint, the shared scenario IDs. The resulting evidence is specific to this simulator and scenario matrix.

Across the five checkpoints, GNN-PPO completed 249/540 routes (46.1%; nested 95% interval 31.3–58.5%), while LSTM-PPO completed 403/540 (74.6%; 58.0–87.6%). The paired LSTM-minus-GNN completion difference was 28.5 percentage points (nested 95% interval 20.6–37.2). Of 540 matched outcomes, both policies succeeded on 242 scenarios, LSTM-PPO alone succeeded on 161, GNN-PPO alone on 7, and both failed on 130. Neither policy proposed an invalid action or used an environment fallback; GNN-PPO encountered 18 no-valid-action states, while LSTM-PPO encountered none. Thus the aggregate advantage is accompanied by substantial shared failure and should not be read as near-universal route feasibility. Figure 4 reports the completion estimate with its nested interval and successful-route hops. Averaging checkpoint-specific successful-hop means gives 87.8 for GNN-PPO and 71.4 for LSTM-PPO; among the 242 scenarios both completed, LSTM-PPO used 32.6 fewer hops on average. Hop count is conditional on success and does not measure route efficiency for failed episodes. Figure 5 resolves completion by season and time; Fig. 6 shows checkpoint variation. These differences describe this evaluation and do not establish generalization beyond its topology and configured channel model.

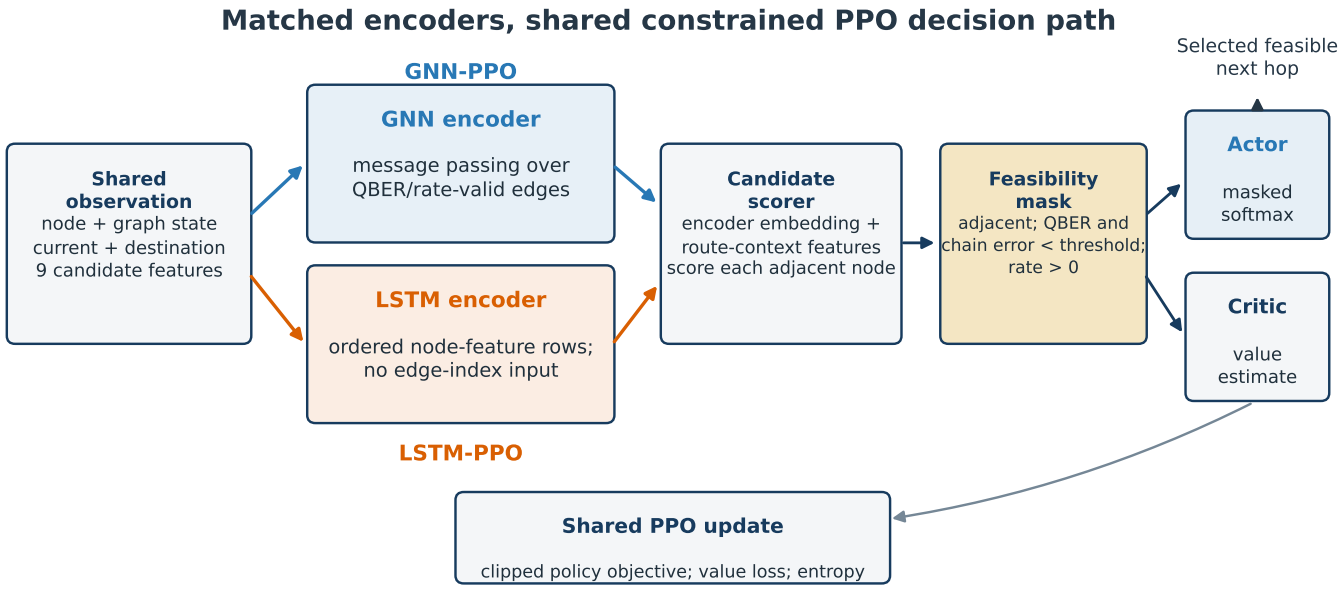


Fig. 3. Matched policy architectures. Both receive the same observation and use the same candidate-scoring, feasibility, actor–critic, and PPO decision path. The GNN encoder aggregates over graph edges; the LSTM encoder processes node-feature rows in a fixed topology order without edge-index input.

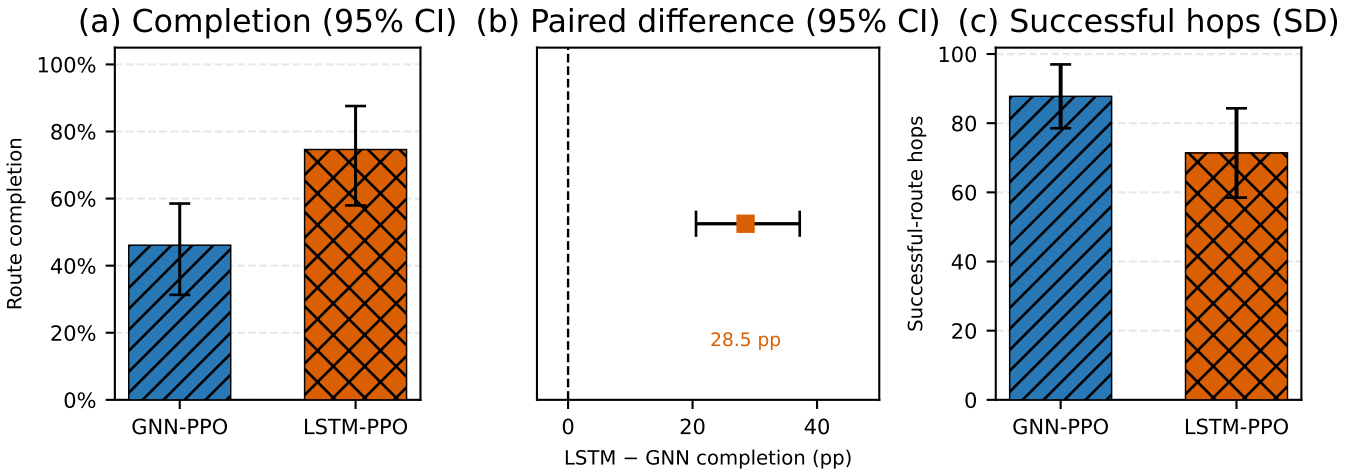


Fig. 4. Matched GNN-PPO and LSTM-PPO results on the strict held-out matrix. (a) Route completion with nested 95% intervals; (b) paired LSTM-minus-GNN completion difference with its nested 95% interval (pp denotes percentage points); and (c) successful-route hops averaged across checkpoint means, with between-checkpoint standard deviation.

Condition-level completion is shown in Fig. 5. The bars summarize means across five checkpoint-specific rates; whiskers show checkpoint standard deviation. LSTM-PPO's mean completion exceeded GNN-PPO's in all six season/time cells, by 24.4–32.2 percentage points. The gaps were not confined to one condition, although the five independent checkpoints per policy remain a small sample for estimating training variability.

Figure 6 presents the five paired checkpoint outcomes. GNN-PPO checkpoint completion ranged from 18.5% to 63.9%; LSTM-PPO ranged from 42.6% to 89.8%. Pairing makes this seed-level variation visible rather than hiding training instability in the aggregate over 540 episodes per architecture.

B. Inference Cost

Actor-call latency was measured on the same CPU host after warm-up, using 1,000 deterministic calls per checkpoint, four CPU threads, and disabled encoder caching. Figure 7 compares the mean checkpoint-specific 95th-percentile latency and its across-checkpoint standard deviation: 9.61±1.65 ms for GNN-PPO and 6.24±0.22 ms for LSTM-PPO. The timing excludes environment simulation and is not end-to-end latency.

C. Reward-Component Audit

A matched reward-component audit evaluates the 20261002 GNN-PPO and LSTM-PPO checkpoints on the same 108 scenario IDs. GNN-PPO completed 50/108 routes (46.3%) with 92.56 mean successful-route hops; LSTM-PPO completed 91/108 (84.3%) with 62.22 hops. This selected-seed

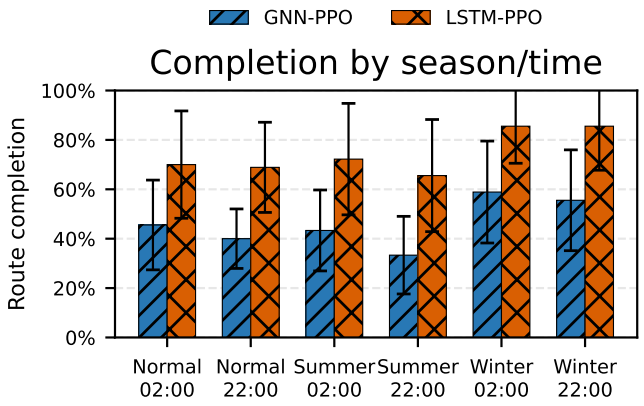


Fig. 5. Held-out route completion by season and time for GNN-PPO and LSTM-PPO. Bars show means across five checkpoints and whiskers show checkpoint standard deviation; colors and hatches distinguish policies.

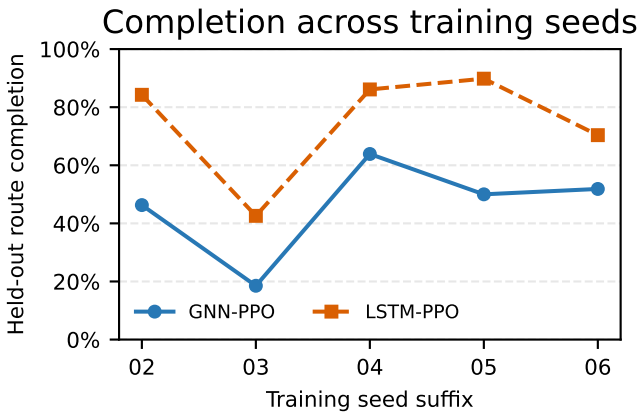


Fig. 6. Held-out route-completion rate for each matched training-seed pair. Seed suffixes identify the five checkpoints; color, marker, and line style distinguish policies.

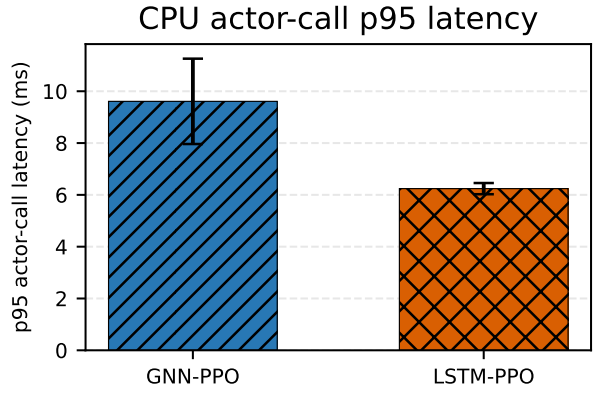


Fig. 7. CPU actor-call p95 latency in milliseconds, averaged across five checkpoints; whiskers show checkpoint standard deviation. Each checkpoint used 1,000 deterministic calls after warm-up, with four CPU threads and encoder caching disabled; environment simulation is excluded.

result is descriptive and is not a second independent estimate of the five-seed comparison. Because component totals accumulate over trajectories of different lengths, they describe the shaped training signal and are not independent measures of delivered key service. Figure 8 plots the component-wise mean difference; it is a diagnostic, not an additional performance objective.

6. DISCUSSION AND LIMITATIONS

A. What the Matched Encoder Comparison Shows

Both methods share the PPO objective and budget, candidate features, feasibility mask, and downstream decision interface. In this held-out matrix, LSTM-PPO had higher route completion, fewer conditional successful-route hops, and lower measured actor-call latency. The comparison is evidence about these two implementations under this simulator's specified conditions, not a general ranking of encoder families.

B. Node-Order Sensitivity and Architectural Confounders

The LSTM reads node-feature rows in a fixed topology order and does not receive the graph's edge index. Its output can therefore depend on an arbitrary serialization of nodes: permuting equivalent node rows can change which features become sequential neighbors even when the underlying graph is unchanged. The hidden state is also reset at each action call, so this model does not use persistent route-history memory. GNN-PPO, by contrast, aggregates using graph connectivity. The shared PPO budget, candidate features, feasibility mask, and downstream interface reduce differences in the decision task, but do not equalize the encoders' inductive biases, effective capacity, or optimization behavior. No node-permutation, parameter-matched, graph-aware sequence, or persistent-memory ablation was run. Accordingly, the observed difference should be attributed to the implemented encoder-policy combinations, not to an intrinsic advantage of LSTMs over GNNs.

C. Training-Seed Variability

The five independently trained checkpoints are the replication units for training variation; the 540 held-out episodes per policy are paired scenario evaluations, not 540 independent training runs. Completion ranged from 18.5% to 63.9% for GNN-PPO and from 42.6% to 89.8% for LSTM-PPO. The nested intervals resample checkpoints and shared scenario IDs, but five checkpoints per architecture provide limited resolution for the tails of the training-seed distribution. The paired estimate is useful for this frozen protocol, while broader claims about run-to-run reliability require more independent training seeds.

D. Classical Comparators and Scope of the Ranking

The current matched study does not establish an advantage over strong classical routing algorithms. A hop-count BFS or a static shortest-path calculation would optimize a different objective and would not by itself represent changing channel availability, feasibility constraints, or key-pool state. A useful classical comparison should use the same topology, route termination rules, edge-feasibility checks, dynamic resource state, and held-out scenarios, with an explicit resource-aware

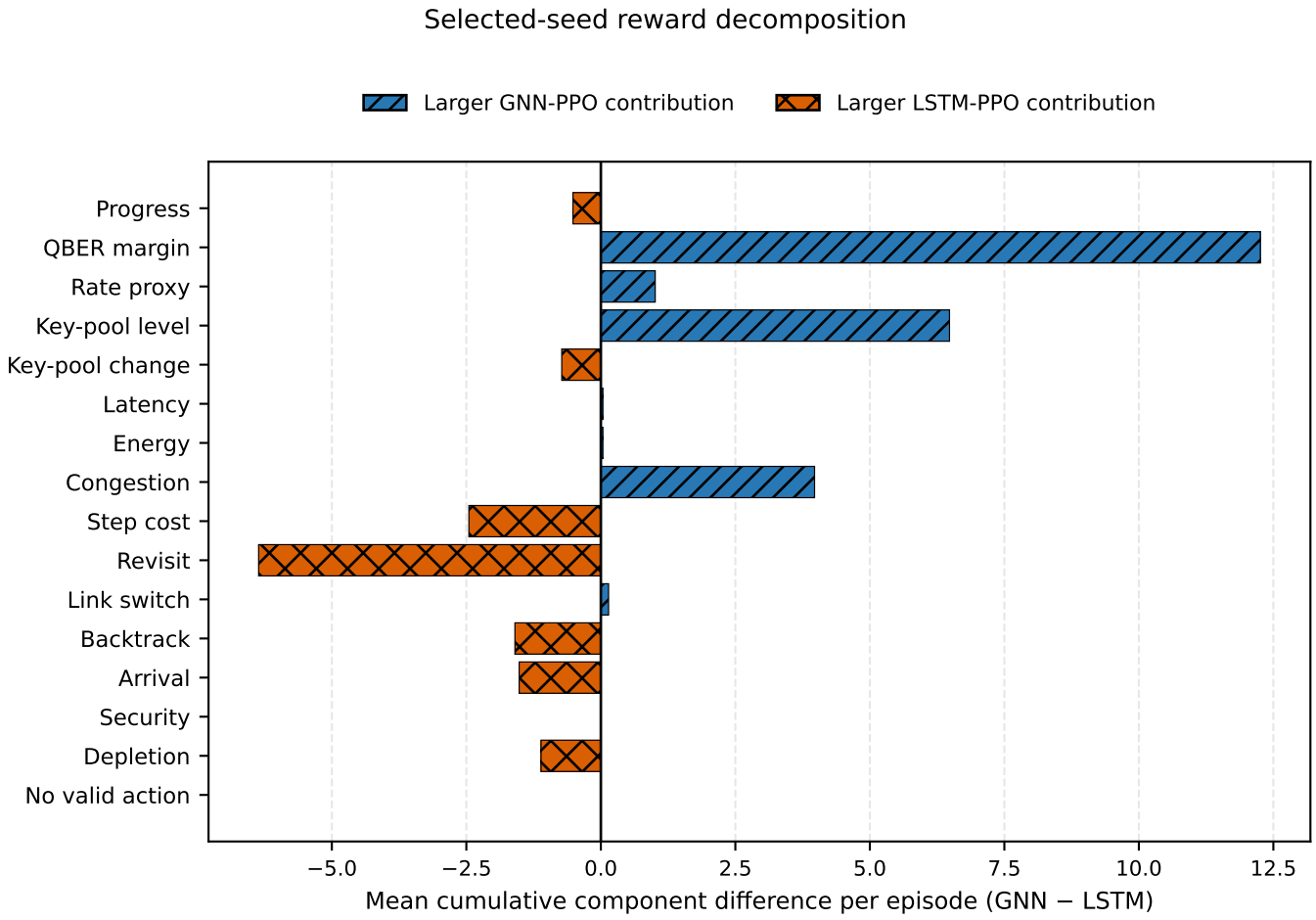


Fig. 8. Mean cumulative reward-component difference for the selected matched checkpoints (GNN-PPO minus LSTM-PPO) on shared scenarios. Blue positive bars indicate a larger GNN-PPO contribution and orange negative bars a larger LSTM-PPO contribution. Because episode lengths differ, this is a descriptive shaped-reward audit, not a service-capacity measure.

cost or state-dependent replanning rule. Until such a baseline is evaluated under the matched environment, the reported ranking is limited to GNN-PPO versus LSTM-PPO.

E. What State Affects the Graph Policy?

Controlled paired-input counterfactuals on a selected GNN-PPO checkpoint hold route context fixed while varying one candidate feature; this is a policy-sensitivity diagnostic, not the GNN-PPO/LSTM-PPO route comparison. Across the tested range, FSO action probability increased from 0.311 to 0.638 as its normalized rate-proxy ratio to fiber rose from 0.25 to 4.0. With QBER below the feasibility threshold, action probability changed only slightly (approximately 0.473 to 0.472); at QBER 0.111 the candidate was masked. In a separate 32-trial turbulence-input sweep, a 100-fold C_n^2 multiplier reduced median FSO rate proxy from 0.202 to 0.183, while mean FSO action probability shifted from 0.749 to 0.740 and FSO remained selected in every trial. The checkpoint is sensitive to proxy-rate contrast and the hard mask, while this particular turbulence scaling had little effect on its local preference. These counterfactual inputs do not establish route-level benefits or calibrated atmospheric response.

F. Model and Network Scope

The archived physical checks give internally consistent outputs for the configured equations: at 80 km and 1550 nm, modeled fiber loss including excess loss is 16.46 dB, detection probability is approximately 0.021 per incident photon, QBER proxy is 0.0105, and normalized rate proxy is 0.0175. The count-rate bound assumes one photon per pulse and is separate from the weak-coherent-pulse finite-key engineering mode. For a 10-km FSO span in the monsoon 22:00 scenario, configured availability is 88% and a 1,000-sample check yielded 88.5%; median normalized rate proxy conditional on availability was 0.2035. These are model checks, not device or field measurements.

The city corridors are OpenStreetMap road proxies, not verified fiber routes; FSO line of sight and weather distributions are uncalibrated, and the fiber and FSO settings do not describe one characterized terminal pair. The key-pool state is normalized and single-route: calibrated key generation/consumption, simultaneous demand, network-wide blocking, and management traffic are not modeled. The normalized rate proxy is not delivered secret-key throughput, and the optional finite-key estimator is not independently established

as composably secure. Thus the channel, resource, routing, and latency results are simulator evidence only. Empirical calibration, multi-demand evaluation, broader topology testing, and the architecture and classical-baseline ablations above remain outside this revision's scope.

G. Separate Small-Graph Reward Oracle Check

The archived exact reward-oracle check uses a static toy graph and a different state/action setting. It is retained in the supplementary research package as a separate diagnostic and is not pooled with, or used to rank policies in, the matched encoder evaluation.

7. CONCLUSION

In this five-seed evaluation of 540 paired scenarios, LSTM-PPO completed 74.6% of routes versus 46.1% for GNN-PPO (difference: 28.5 percentage points; nested 95% interval: 20.6–37.2). This supports the tested LSTM implementation only within this simulator and scenario matrix; it is not a general architecture ranking. Network and channel values remain uncalibrated proxies.

Open questions include the effects of node permutations, parameter-matched and graph-aware sequence encoders, persistent LSTM state, more training seeds, and broader topologies. Calibrated channel and key-pool dynamics and classical resource-aware baselines under the same held-out conditions are needed before assessing field performance or delivered secret-key service.

ACKNOWLEDGMENT

Funding and institutional acknowledgments are to be supplied by the authors.

DATA AND CODE AVAILABILITY

The simulator code, archived configurations, scenario outputs, and generated figures are included with the supplementary research package. The road-corridor geometries derive from OpenStreetMap data and are provided under the Open Database License. Publications using the map should credit "Map data from OpenStreetMap contributors" and link to https://www.openstreetmap.org/copyright. The geometries are routing proxies and do not represent verified fiber infrastructure.

REFERENCES

[1] V. Scarani, H. Bechmann-Pasquinucci, N. J. Cerf, M. Dušek, N. Lütkenhaus, and M. Peev, "The security of practical quantum key distribution," *Reviews of Modern Physics*, vol. 81, no. 3, pp. 1301–1350, 2009.

[2] W. J. Munro, K. Azuma, K. Tamaki, and K. Nemoto, "Inside quantum repeaters," *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 21, no. 3, p. 6400813, 2015.

[3] "ITU-T Recommendation Y.3800: Overview on networks supporting quantum key distribution," International Telecommunication Union, Geneva, Switzerland, 2019.

[4] "ITU-T Recommendation Y.3803: Key management for quantum key distribution networks," International Telecommunication Union, Geneva, Switzerland, 2020.

[5] M. Mehic, M. Niemiec, S. Rass, J. Ma, M. Peev, A. Aguado, V. Martin, S. Schauer, A. Poppe, C. Pacher, and M. Voznak, "Quantum key distribution: A networking perspective," *ACM Computing Surveys*, vol. 53, no. 5, pp. 101:1–101:41, 2020.

[6] T. Johann, M. Wenning, D. Giemsa, A. Dochhan, M. Gunkel, T. Fehenberger, and S. Pachnicke, "Comparison and optimization of different routing methods for meshed qkd networks using trusted nodes," *Journal of Optical Communications and Networking*, vol. 16, no. 3, pp. 382–391, 2024.

[7] P. Sharma, S. Gupta, V. Bhatia, and S. Prakash, "Deep reinforcement learning-based routing and resource assignment in quantum key distribution-secured optical networks," *IET Quantum Communication*, vol. 4, no. 3, pp. 136–145, 2023.

[8] J. Roik, K. Bartkiewicz, A. Černoch, and K. Lemr, "Routing in quantum communication networks using reinforcement machine learning," *Quantum Information Processing*, vol. 23, no. 89, 2024.

[9] O. Amer, W. O. Krawec, V. U. Manfredi, and B. Wang, "Dynamic routing for quantum key distribution networks," *arXiv preprint arXiv:2212.03144*, 2023.

[10] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," in *Proceedings of the IEEE International Conference on Computers, Systems and Signal Processing*, Bangalore, India, 1984, pp. 175–179.

[11] W.-Y. Hwang, "Quantum key distribution with high loss: Toward global secure communication," *Physical Review Letters*, vol. 91, no. 5, p. 057901, 2003.

[12] H.-K. Lo, X. Ma, and K. Chen, "Decoy state quantum key distribution," *Physical Review Letters*, vol. 94, no. 23, p. 230504, 2005.

[13] D. Gottesman, H.-K. Lo, N. Lütkenhaus, and J. Preskill, "Security of quantum key distribution with imperfect devices," *Quantum Information and Computation*, vol. 4, no. 5, pp. 325–360, 2004.

[14] M. Tomamichel, C. C. W. Lim, N. Gisin, and R. Renner, "Tight finite-key analysis for quantum cryptography," *Nature Communications*, vol. 3, p. 634, 2012.

[15] C. C. W. Lim, M. Curty, N. Walenta, F. Xu, and H. Zbinden, "Concise security bounds for practical decoy-state quantum key distribution," *Physical Review A*, vol. 89, no. 2, p. 022307, 2014.

[16] W. O. Krawec, B. Wang, and R. Brown, "Finite key security of simplified trusted node networks," in *Proceedings of the IEEE International Conference on Quantum Computing and Engineering*, 2024, pp. 1777–1787.

[17] *ITU-R Recommendation P.1814-1: Prediction Methods Required for the Design of Terrestrial Free-Space Optical Links*, International Telecommunication Union Std., 2025.

[18] A. Kržič, S. Sharma, C. Spiess, U. Chandrashekara, S. Töpfer, G. Sauer, L. J. G.-M. del Campo, T. Kopf, S. Petscharnig, T. Grafenauer, R. Lieger, B. Ömer, C. Pacher, R. Berlich, T. Peschel, C. Damm, S. Risse, M. Goy, D. Rieländer, A. Tünnermann, and F. Steinlechner, "Towards metropolitan free-space quantum networks," *npj Quantum Information*, vol. 9, p. 95, 2023.

[19] E. W. Dijkstra, "A note on two problems in connexion with graphs," *Numerische Mathematik*, vol. 1, pp. 269–271, 1959.

[20] R. Bellman, *Dynamic Programming*. Princeton, NJ, USA: Princeton University Press, 1957.

[21] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[22] J. Schulman, P. Moritz, S. Levine, M. I. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," in *Proceedings of the International Conference on Learning Representations*, 2016.

[23] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra, "Planning and acting in partially observable stochastic domains," *Artificial Intelligence*, vol. 101, no. 1–2, pp. 99–134, 1998.

[24] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

[25] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proceedings of the International Conference on Learning Representations*, 2017.

[26] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 1024–1034.

[27] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proceedings of the International Conference on Learning Representations*, 2018.

[28] M. Haklay and P. Weber, "Openstreetmap: User-generated street maps," *IEEE Pervasive Computing*, vol. 7, no. 4, pp. 12–18, 2008.

[29] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.